

Windows CMD Administration Commands

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1. File & Directory Management

- **dir /a** : List all files and directories, including hidden and system files (similar to **ls -la**).
- **cd** : Print the current working directory (similar to **pwd**). Use **cd \path** to change directories.
- **robocopy /source /dest /mir** : The ultimate file copy tool. The **/mir** flag mirrors a directory tree perfectly. (*Expert tip: Always use **robocopy** over the legacy **copy** or **xcopy** commands for large transfers*).
- **move source.txt dest.txt** : Move or rename a file or directory.
- **del /f /q filename** : Forcefully (**/f**) and quietly (**/q**) delete a file.
- **rmdir /s /q directory_name** : Recursively remove a directory and all its contents without prompting (the Windows equivalent of **rm -rf**).
- **findstr /i "pattern" filename** : Search for a text string inside a file, ignoring case (**/i**) (the CMD equivalent of **grep**).

2. System, Memory, & Storage

- **systeminfo** : Displays detailed configuration information about the computer and its operating system, including uptime and installed hotfixes.
- **tasklist** : Displays a list of currently running processes (similar to **ps**).
- **chkdsk C: /f /r** : Scans the file system for logical and physical errors. **/f** fixes errors, **/r** recovers readable information from bad sectors.
- **fsutil volume diskfree C:** : Quickly check the total and available free space on a drive.
- **wmic memorychip get capacity, speed** : Query the Windows Management Instrumentation (WMI) to get physical RAM details.
- **sfc /scannow** : The System File Checker. It scans and repairs corrupted native Windows system files.

3. Networking

- **ipconfig /all** : Display detailed IP address, DNS, and MAC address information for all network adapters.
- **ipconfig /flushdns** : Clear the DNS resolver cache (essential for troubleshooting stale DNS entries).
- **route print** : Display the local routing table.

- **ping -t google.com** : Send continuous ICMP echo requests until manually stopped with **Ctrl+C**.
- **tracert google.com** : Trace the network path and routing hops to a destination.
- **netstat -ano** : Show all active TCP connections, listening ports, and the specific Process ID (PID) owning them (crucial for finding port conflicts).
- **nslookup domain.com** : Query DNS records for a specific domain.

4. User & Group Management

- **whoami /user /groups** : Display your current username, Security Identifier (SID), and all groups your account belongs to.
- **net user username /add** : Create a new local user account.
- **net user username newpassword** : Reset or change a user's password.
- **net localgroup Administrators username /add** : Add a specific user to the local Administrators group.
- **net accounts** : View the local password and lockout policy settings.

5. Process & Service Management

- **taskkill /f /pid <PID>** : Forcefully (/f) kill a specific process by its ID (the Windows equivalent of **kill -9**).
- **taskkill /f /im notepad.exe** : Forcefully kill a process by its image/executable name.
- **sc query service_name** : Check the exact status of a Windows service.
- **net start service_name** : Start a stopped service (e.g., **net start w3svc** for IIS).
- **net stop service_name** : Stop a running service.

6. Permissions & Ownership (NTFS)

- **icacls "C:\path" /grant Username:(OI)(CI)F** : Grant a user Full Control (F) over a folder, and ensure those permissions inherit down to all Objects (OI) and Containers (CI) inside it.
- **icacls "C:\path" /reset /t /c** : Strip customized permissions and reset the folder (and subfolders /t) to inherit the default permissions from its parent.
- **takeown /f "C:\path" /r /d y** : Forcefully take ownership of a file or directory and all its contents (/r), automatically answering yes (/d y) to prompts.

7. Active Directory & Domain Basics

- **gpupdate /force** : Force an immediate background refresh of all Group Policy Objects (GPOs) applied to the machine and user.
- **gpresult /r** : Generate a quick summary report of which Group Policies are actually being applied to the machine and user.

- **set logonserver** : Quickly check which specific Domain Controller authenticated your current session.

Pro-Tip for CMD:

If you need to chain commands together conditionally in CMD (like `&&` in Linux), you can do so! For example: `ipconfig /release && ipconfig /renew` will only attempt to renew if the release is successful.

To view the manual for any CMD command, simply append `/?` (e.g., `robocopy /?` or `icacls /?`).

CMD